

Comparative Studies of Six Programming Languages

Abstract

Comparison of programming languages is a common topic of discussion among software engineers. Multiple programming languages are designed, specified, and implemented every year in order to keep up with the changing programming paradigms, hardware evolution, etc. In this paper we present a comparative study between six programming languages: C++, PHP, C#, Java, Python, VB; these languages are compared under the characteristics of reusability, reliability, portability, availability of compilers and tools, readability, efficiency, familiarity and expressiveness.

1 Introduction

Programming languages are a fascinating and interesting field of study. Computer scientists tend to create new programming languages. Thousands of different languages have been created in the last few years. Some languages enjoy wide popularity and others introduce new features. Each language has its advantages and drawbacks. The present work provides a comparison of various properties, paradigms, and features used by a couple of popular programming languages: C++, PHP, C#, Java, Python, and VB. With these languages and their widespread use, software designers and programmers should be aware of the benefits and drawbacks that each language could bring to their software solutions and be careful when they make rational decisions. These languages are compared under the characteristics of reusability, reliability, portability, availability of compilers and tools, readability, efficiency, familiarity, and expressiveness. Other criteria like the programming effort, run time efficiency, memory consumption, and database connectivity are disclosed by implementing and running the same set of programs using all the languages under study.

1.1 Related Work

In this comparative study, relevant research has been made based on trusted websites, research papers, scientific articles, and textbooks.

1.2 Overview

The rest of the paper is organized as follows. First we introduce the programming languages under study and we introduce their development history including their important contributors, versions,

and paradigms. Then we address how each language is evaluated, typing strategies used, and how memory is managed. Then we highlight the strengths and weaknesses for each language as well as the application domains in which these languages are mainly used. A couple of experiments will be conducted to compare efficiency, GUI development, and database connectivity followed by analysis and results interpretations. Moreover, we will compare language features and GUI development facilities.

2 Programming Languages

2.1 PHP

2.1.1 History and Versions

PHP (Hypertext Pre-processor) is a server-side scripting language designed for web development. It is a powerful language that runs on more than 244 million websites. Rasmus Lerdorf first created PHP in 1994. He wrote a series of Perl scripts used to enhance his personal website. The main objective to create these scripts was to enhance his personal website and improve its performance. Another reason to create PHP was to monitor his online resume and related information. Later on, Rasmus started to develop C scripts to add the ability to work with web forms and access databases. One year later, Rasmus combined PHP with his own form interpreter and then released PHP 2.0 on June 8, 1995. Two years later, two programmers, Zeev Suraski and Andi Gutmans, released PHP/FI in 1997 by rebuilding the PHP core system. In 1998, these programmers ported many PHP/FI utility tools into PHP 3, which was released in 1998 to support multiple platforms, web servers, and a wide number of databases. This PHP 3 version was the first widely used version at that time. PHP 4 was developed by the PHP development team and Zend Technologies to improve speed and reliability over PHP 3 and then released in May 2000. In fact, PHP 4 introduced new features to the language such as reference and Boolean support, COM support on Windows, output buffering, and expanded object-oriented programming. Finally, PHP 5 was introduced in July 2004. Powered by Zend Engine II, PHP 5 was an improved version of PHP 4 and included many features such as improved support for object-oriented programming and provided a well-defined and consistent PDO interface to access databases. Another important feature implemented in PHP 5 is backward compatibility with earlier versions of PHP. Late binding was added to later versions of PHP 5 (v5.3). Some other significant variants between PHP 5 and PHP 4 are:

- Introducing interfaces.
- Incorporating static methods with properties.
- Allowing classes to be declared as **final**.
- Introducing three levels of visibility: public, private, and protected.
- Adding support for exceptions.

PHP takes most of its language from C, Java, and Perl. Similarly to Java and C++, PHP has native Boolean types. Also, in terms of keyword and syntax, PHP has similar keywords to most

high-level languages that follow the C style. Similarly to C, PHP stores whole numbers in a platform-dependent range: 32- or 64-bit signed integers. PHP borrows the `if`, `while` loops, and function return style from C and Java. Moreover, it uses most C built-in data types. In fact, PHP has not influenced the development of any programming language to date. However, there are successful and complex tools and software systems in the market that are mainly written in PHP, such as Joomla CMS and Facebook.

2.1.2 Evaluation Strategy

Compilation and execution. PHP scripts are interpreted in the web server. PHP script is first translated into machine language: object code. Then the linker links this file and sends the results back to the client machine.

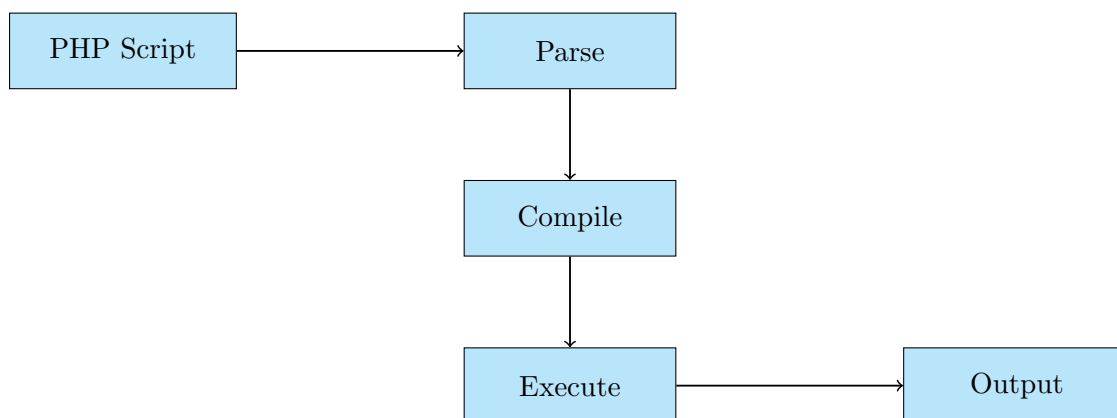


Figure 1: PHP program evaluation process

Unlike a compiled language like C++ or C, PHP code is not compiled directly to machine code. In fact, it is compiled to an intermediary bytecode which is later interpreted by the PHP runtime engine. This evaluation process allows PHP to be easily, quickly and efficiently ported to several different platforms.

Unlike C++, PHP gets evaluated by an interpreter. Both convert a high level language into a binary form that is recognized by computer hardware. Unlike the C++ compiler, the PHP interpreter translates one statement at a time and executes it, while the C++ compiler translates the entire program at once and then executes it. As a matter of fact, C++ takes more time to get analyzed and processed comparatively to analyzing and processing PHP code. As a result, the overall execution time of C++ compiled code is faster than PHP interpreted code.

Typing strategies. PHP is a dynamically typed and weak language. PHP is dynamic since its variables do not need to be defined before they are used. This implies that PHP does not require any explicit declaration of a variable before being used. PHP is a weakly typed language since its variables are not bound to a specific data type. As an example:

```

/* PHP code */
<?php

```

```

$foo = "x";
$foo = $foo + 2; // not an error
echo $foo;
?>

```

In this example, `foo` is first declared as a string type. Then we add 2 to the string. This is allowed in PHP as a part of weak typing.

Memory management. Memory management is critical to PHP programs. The first reason to implement garbage collection for PHP is to reduce memory usage by releasing memory and cleaning up circular-referenced variables. PHP garbage collection uses reference counting to manage its internal variables. This technique is used to keep track of the stored references to memory locations held by objects. When an object is destroyed, all references to this object are removed. However, reference counting fails to deal with circular references. Cycle collection handles this issue where memory cleanup is invoked automatically when the root buffer is full or `gc_collect_cycles()` is called. However, running memory cleanups by the PHP garbage collector can introduce performance decline and run time slowdowns when cycle collection runs.

2.2 Visual Basic

2.2.1 History and Versions

Visual Basic was first designed by mathematician professors John Kemeny and Thomas Kurtz at Dartmouth College in New Hampshire in the early 1960s. The main idea to design this new language was to make it easy for college students to learn and use. The language first started as a part of FORTRAN but eventually they started creating a new language called BASIC. This BASIC language became so popular because it was easy to use. In the mid 1970s–1980s it became so popular that it started shipping with microcomputers. This allowed small business owners and professionals to develop customized software at affordable price.

General Electric used BASIC to design their largest time sharing system. In the mid 1970s, Bill Gates and Paul Allen wrote one of the versions of BASIC for the first microcomputers called the MITS Altair. The introduction of BASIC in Microsoft Windows operating systems made a new milestone for the BASIC language. In 1991, VB 1.0 was released as a combination of code “Ruby” and “Embedded Basic.” Ruby was developed by Alan Cooper, who is known as the father of Visual Basic. In 1987, Alan Cooper got an idea while interviewing a client who wanted a wide range of shell solutions. So he decided to start creating a language named “Tripod,” a set of tools where each user could customize their own application. During development of Tripod, Alan came up with the idea of drag and drop. In 1988, Alan showed his Tripod prototype to Bill Gates, who bought it. After that, a team of skilled programmers started to work on a stable release of Tripod, now called Ruby. In 1990, after 18 months, the language was ready to release and was tested by Microsoft’s quality team. The original intention of shipping Ruby with Windows 3.0 did not happen; Microsoft delayed shipping. Microsoft converted it from shell scripting to visual programming by adding QuickBasic. In 1991 Microsoft released VB 1.0. This language brought evolution to high-level languages. Visual Basic eventually kept evolving and nowadays it is a main part of the .NET framework. Also, other languages like C# are highly influenced by Visual Basic.

Versions. There are several versions of Visual Basic as it evolved.

Visual Basic Learning Edition: It includes basic features of Visual Basic and allows programmers to create powerful applications for Windows.

Visual Basic Professional Edition: It allows professionals to develop solutions using the full featured set of tools. It includes all features of the Learning Edition and includes ActiveX controls, Internet Information Server application designer, and Dynamic HTML Page designer.

Visual Basic Enterprise Edition: It allows creating robust distributed applications. It has all professional features. Also, it has support for backend tools such as SQL Server, Microsoft Transaction Server, Internet Information Server, and Visual SourceSafe.

Furthermore Visual Basic evolved and in 1998 Visual Basic 6 was released. VB6 was the stable version of Visual Basic and became more popular. After that, Microsoft started creating the .NET framework and with the first version of .NET they released a new version of Visual Basic known as Visual Basic .NET. Over time, VB has known enhancements, and Visual Basic 2013 was current in .NET Framework 4.5.

2.2.2 Paradigms

Visual Basic mainly uses two programming paradigms: object-based and object-oriented. Initially, Visual Basic was an object-based programming language. That means that it has notions of objects, classes, data abstraction, and strong type checking, but it does not support inheritance, subtyping, and polymorphism. Visual Basic 6 was object-based. As the language evolved, it transformed into an object-oriented language. Visual Basic evolved through the decade it grew as a structured programming language in the 1970s to make programs easier to understand, develop and maintain.

Programming motivates the use of modularization where a developer can divide a large program into smaller parts. This creates functions that can be reused several times within the project. Visual Basic was object-based and very popular at that time. Eventually, object-oriented concepts became more popular and Visual Basic needed to get updated. As a result, Microsoft released a new object-oriented version of Visual Basic known as Visual Basic .NET. Visual Basic evolved from COM programming to CLR compilation. Visual Basic was first static and strong type checking but as it evolved the type checking also evolved and Visual Basic .NET uses both strong and weak type checking. Types of various constructs like variables and function expressions can be checked to reduce bugs in programs and this can be done statically at compile time and dynamically at run time. Visual Basic .NET is capable of both.

The most important programming paradigm of Visual Basic is event-driven. The basic concept of event-driven is that it partially allows programming logic that depends on user interaction to control program execution. The user triggers the event by clicking on either graphical controls or keyboard keys. Each event has its own event handler that is related with programming code. Event-driven programming is preferred by most users as the events are determined by end users, which allows them to interact with the application. It is widely used in graphical user interface design.

2.2.3 Strengths, Weaknesses, and Application Areas

Application area. Visual Basic is now a part of the .NET framework, so the developer can use VB.NET to create Windows, web, and console applications. Developers can create large, powerful, high-quality applications using VB.NET. Depending on their needs, programmers can choose one of the options mentioned above to develop the application. VB.NET is widely used for in-house applications. It can also be used to create AJAX and COM components to use either online or on desktop applications.

Advantages. VB.NET provides rich GUI support and OOP concepts which makes it a competitive language to develop interactive applications. VB.NET has a modern interface style which gives a good and simple look to developers. It uses static type checking which analyzes the source code and gives errors like immediate syntax errors, such as missing characters and using wrong keywords. It has the ability to compile and run within the IDE. It has customization capabilities which allow users to highly customize IDE appearance. One of the interesting advantages of VB is its simplicity. It is an easy-to-learn language. VB is a component integration language which utilizes Microsoft's component object model that allows any system part to be bolted onto a program easily.

Disadvantages. VB.NET generates lots of automated code which makes it difficult for developers to understand. Visual Basic is a proprietary programming language written by Microsoft, so programs written in Visual Basic cannot easily transfer to another language. Visual Basic is part of the .NET framework which has many other languages. So, VB loses competitiveness in front of Microsoft languages like C#, since they all share the same framework and they can all perform the same tasks.

2.2.4 Compilation and Execution

Visual Basic is part of the .NET framework so it compiles and executes in the same manner as other languages that run on .NET, like C#.

As shown by the .NET framework, code written in Visual Basic first gets converted into Microsoft intermediate language. This code gets transferred to the CLR (Common Language Runtime) and then converted into machine native code.

2.2.5 Memory Management

Visual Basic has a garbage collector similar to Java and C# which automatically manages the memory of programs. The garbage collector runs automatically and checks for unused objects and releases their memory. When a developer creates an object in C#, memory is automatically allocated. But if the object is no longer used, the garbage collector releases the memory. However, if a developer wants to manage memory manually, they can use `GlobalAlloc` to allocate memory and `GlobalFree` to release it.

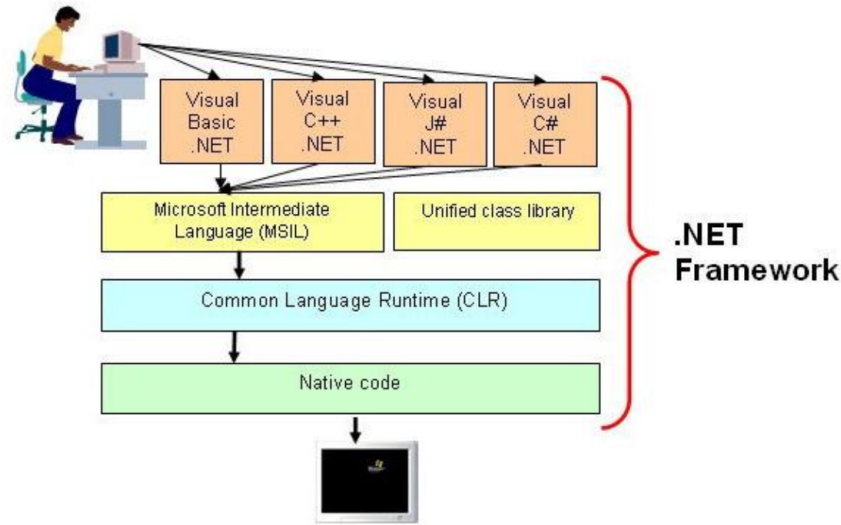


Figure 2: .NET framework compilation and execution flow

2.2.6 Typing Strategy

Visual Basic is a statically type-checked language. It verifies type safety of the program by analyzing the source code. This makes the program less faulty and allows many bugs to be caught early in the development phase. In general, the developer gets errors at compile time so they can solve them early, which saves debugging time and makes the program less error-prone. Visual Basic is strongly typed. It focuses more on the type of a variable during compile time. For example, the type of a calling function object must be the same type as the called function's object.

2.2.7 Available Compilers and IDEs

Since VB.NET is a part of the .NET framework, it comes with the best IDE developed by Microsoft so far: Visual Studio. Visual Studio is a leading IDE for VB.NET. It provides a rich GUI interface, custom controls, a built-in compiler, and integration with IIS, so users can press start to run applications without worrying about other details. Visual Studio comes with different versions with different commercial solutions and a free version.

There is also SharpDevelop, a free and open source development environment for the .NET platform. SharpDevelop has similar features to Visual Studio. It has syntax highlighting, auto-completion, and toolbar features. It uses GUI technologies like Windows Forms, Windows Presentation Foundation, and Entity Framework. It has many versions like SharpDevelop 5, 4.4, and 3.2.